



Quasi-average predictions and regression to the trend: An application to the M6 financial forecasting competition[☆]

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ABSTRACT

This paper presents the winning method that achieved fifth place overall in the M6 financial forecasting competition. The method is based on the idea that, under the efficient market hypothesis, it is often more effective to predict values close to the expected averages of categories and trends than to try to make precise predictions. By leveraging low-variability prediction methods, we forecast both the relative performance of multiple assets and their optimal investment positions. We demonstrate that combining asset-class and temporal averages yields modest but consistent advantages over reference indices. The results highlight the challenges of achieving above-average returns in efficient markets and the potential benefits of low-variability prediction methods in such contexts.

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1. Introduction

The efficient market hypothesis regards financial markets as informationally efficient, with all available information already reflected in asset prices (Fama, 1970; Malkiel, 2019), implying that it is impossible to achieve above-average returns consistently by trading on publicly available information, as any new information is rapidly incorporated into stock prices. Consequently, this hypothesis challenges the notion of consistently outperforming the market and renders most forms of fundamental and technical analysis eventually futile (Samuelson, 1974).

Indeed, over the past 10 and 20 years, less than 7% and 10%, respectively, of U.S. actively managed equity funds have outperformed their benchmarks (Edwards, Ganti, Lazzara, Nelesen, & Di Gioia, 2023). Since the conclusion of 2022, the most recent instance when the average active

U.S. stock fund outperformed the S&P 500 stock index for an entire calendar year was in 2009.

This type of underperformance was also highlighted in a study of all 2132 actively U.S. managed funds, except for those concentrating on niche market segments or employing leverage, from June 2017 to June 2022 (Lazzara, Stoddart, & Di Gioia, 2023), which found that none of the funds maintained a top-quarter performance ranking each year over the entire five-year period. Relaxing the criterion to a top-half performance in each of the five years under examination resulted in less than 1% of all funds doing so. Remarkably, this value is lower than the 6.25% expected persistence under random distribution (the probability of the top half remaining on the top for the next four years is $0.5^4 = 0.0625$). This effect indicates that high performers tend to perform worse than random later on, which would be consistent with a regression to the mean effect (Stigler, 1997).

Therefore, forecasting methods, along with the efficacy of various trading strategies, are severely challenged by efficient markets, as prices are already a reflection of all available information. However, there can potentially be some signals from the inherent asset price dynamics

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and economic indicators evolution that could be used. These challenges and potential signals further raise the question of whether indeed actively managed portfolios can consistently outperform passive index funds in the long run.

The M6 Financial Forecasting Competition is the sixth installment of the Makridakis competitions (Makridakis, Fry, Petropoulos, & Spiliotis, 2022; Makridakis et al., 2023). Its focus on forecasts of stock returns and risk lies at the center of the efficient market hypothesis. It asked participants to forecast the relative performance of multiple assets and their investment positions from a universe of 50 S&P500 stocks and 50 international exchange-traded funds (ETFs), encompassing diverse categories and geographical areas. The relative performance track consisted of forecasting the rank of each asset returns. The investment position track consisted of developing a portfolio, with potentially long and short positions, to obtain maximum returns adjusted by the risk. The M6 competition spanned 12 submission periods of four weeks each. Remarkably, while previous M-competitions, particularly M4 and M5, were dominated by sophisticated machine learning approaches (Montero-Manso, Athanasopoulos, Hyndman, & Talagala, 2020), the M6 competition demonstrated that computationally simpler methods, such as the approach presented here, could achieve excellent performance in financial forecasting contexts (Makridakis et al., 2023).

Here, we present an approach that outperformed the benchmark indices in both the relative performance and the investment positions and that ranked fifth overall in their combined score in the M6 competition. The approach is based on the idea that, in high-variability contexts, it is often more effective to predict values close to the expected averages of categories and trends than to try to make precise predictions. The approach considered only data within the M6 competition universe, namely, the historical values of the 100 assets. While variability is a well-known characteristic of stock prices, often referred to as volatility, here we focus on forecast variability. This quantity is crucial for understanding the reliability of prediction methods, as high forecast variability can lead to less consistent performance. By reducing forecast variability, we aim to improve the overall accuracy and robustness of our forecasting approach.

Explicitly, the focus is on methods that combine average values. For forecasting performance, we considered the average performance distribution of stocks and ETFs separately and combined it with the average performance distribution over time for each specific asset. The motivation stems from the property of the average value as the quantity that minimizes the mean squared error of a set of random variables. Substantial departures from the average value would not be beneficial unless they are highly informative. For the investment decisions, the strategy consisted of selecting the stocks that performed better in the long term but not too well in the short term. Discarding short-term overachievers is intended to remove recent fluctuations that would potentially regress to the trend. To further suppress the variability resulting from the reference index, we shorted ETFs. The motivation for this type of shorting is that, although it would reduce the returns, it would also reduce the volatility, which leads to a higher information ratio for historical data.

2. Forecasting performance

The forecasting performance was measured by the ranked probability score (RPS), as detailed in Makridakis et al. (2023). In brief, the RPS for asset i in period T is calculated as

$$RPS_{i,T} = \frac{1}{5} \sum_{j=1}^5 \left(\sum_{k=1}^j q_{i,T,k} - \sum_{k=1}^j f_{i,T,k} \right)^2. \quad (1)$$

Here, the ranking of the return of asset i at time T is described by a vector $q_{i,T}$ with components $q_{i,T,k}$ for $k \in 1, \dots, 5$, which are set equal to one if the asset is ranked in the quintile k and zero otherwise. The forecasted probabilities of each rank for a particular asset are denoted by $f_{i,T,k}$. The overall RPS for multiple submission points from T_1 to T_2 is

$$RPS_{T_1:T_2} = \frac{1}{100(T_2 - T_1 + 1)} \sum_{T=T_1}^{T_2} \sum_{i=1}^{100} RPS_{i,T}. \quad (2)$$

The RPS is non-negative in general and reaches zero for a perfect score. The method that relies on the least information, used as a forecasting benchmark, considers the probability of each quintile being realized equal to 0.2 for all assets (i.e., each asset is equally likely to have a performance in any of the quintiles). It leads to an RPS equal to 0.16. It is used as a benchmark because it is consistent with the efficient market hypothesis and is difficult to improve by a large margin. Other naïve approaches perform substantially worse. For instance, predicting only the first, second, third, fourth, or fifth quintile with probability one for all the assets would lead to a RPS of 0.4, 0.28, 0.24, 0.28, and 0.4, respectively, all of which are higher than the benchmark. Randomly predicting any quintile with probability 1 would lead on average to an RPS of 0.32. These naïve approaches also indicate that predicting further away from the expected averages worsens the forecast performances in high-variability scenarios.

The way variability affects nonlinear systems is complex and often paradoxical (Dima & Dima, 2024; Hansen & Singleton, 1982; Vilar & Rubi, 2001). To gain insights into the effects of high variability, it is useful to investigate the conditions for optimality of the average performance. Below, throughout the paper, we use angle brackets $\langle \rangle$ to denote expected values over the possible realizations of the random variables, conditional on the information available at time T . At time T , the information set includes both the realizations $q_{i,T,j}$ and their forecasts $f_{i,T,j}$. While the forecasts can typically rely only on performance historical data and any other relevant information available at the time of producing the forecast, they are part of the known information when the expected RPS is evaluated at time T . We first consider the expected value of the RPS with respect to the variables $q_{i,T,j}$ for given $f_{i,T,j}$, which is expressed as

$$\langle RPS_{i,T} \rangle = \frac{1}{5} \sum_{j=1}^5 (\langle Q_{i,T,j} \rangle - F_{i,T,j})^2 + \frac{1}{5} \sum_{j=1}^5 \text{Var}(Q_{i,T,j}), \quad (3)$$

with $Q_{i,T,j} = \sum_{k=1}^j q_{i,T,k}$ and $F_{i,T,j} = \sum_{k=1}^j f_{i,T,k}$. Here, $\text{Var}(Q_{i,T,j})$ represents the variance of $Q_{i,T,j}$. Note that

$\text{Var}(Q_{i,T,j})$ represents the variance of $Q_{i,T,j}$ over its possible realizations. Since $q_{i,T,j}$ is independent of $f_{i,T,j}$, this variance term captures the inherent variability of the ranks, regardless of the forecasts. The condition of extremum is

$$\frac{\partial}{\partial F_{i,T,j}} \langle RPS_{i,T} \rangle = -\frac{2}{5} (\langle Q_{i,T,j} \rangle - F_{i,T,j}) = 0, \quad (4)$$

which implies that $F_{i,T,j} = \langle Q_{i,T,j} \rangle$ and equivalently $f_{i,T,k} = \langle q_{i,T,k} \rangle$. The extremum is a minimum because $\frac{\partial^2}{\partial F_{i,T,j}^2} RPS_{i,T} = \frac{2}{5}$ is positive.

The estimation of $\langle q_{i,T,k} \rangle$, however, is challenging because of constantly changing market conditions. Low-variability approaches, which reduce fluctuations in the forecast, are generally expected to perform better on average because the evaluation metric (RPS) introduces a nonlinearity. Specifically, the evaluation metric behaves as a convex function, meaning that minimizing forecast variability reduces the risk of large errors disproportionately impacting performance. This result is consistent with Jensen's inequality, which states that for convex functions, the expectation of the function's value is minimized when it is applied to average values rather than variable ones (Grinblatt & Linnainmaa, 2011; Vilar & Rubi, 2018). The requirements can be clarified by considering the forecast variability as $e_{j,T,k} = F_{j,T,k} - \langle F_{j,T,k} \rangle$, which is zero on average by definition. Using the identity $F_{j,T,k} = \langle F_{j,T,k} \rangle + e_{j,T,k}$ in the expression for $RPS_{i,T}$ allows us to rewrite its expected value as

$$\begin{aligned} \langle RPS_{i,T} \rangle &= \frac{1}{5} \sum_{j=1}^5 (\langle Q_{i,T,j} \rangle - \langle F_{i,T,j} \rangle)^2 - \frac{2}{5} \sum_{j=1}^5 \langle Q_{i,T,j} e_{i,T,j} \rangle \\ &\quad + \frac{1}{5} \sum_{j=1}^5 \langle e_{i,T,j}^2 \rangle + \frac{1}{5} \sum_{j=1}^5 \text{Var}(Q_{i,T,j}). \end{aligned} \quad (5)$$

This equation, which is equivalent to Eq. (3), indicates that to enhance the forecasting performance (lower the expected RPS), the variability in the forecast needs to be highly correlated with the rank of the realized returns. In other words, the term $\sum_{j=1}^5 \langle Q_{i,T,j} e_{i,T,j} \rangle$ must be sufficiently large to offset the positive contribution of the forecast variability term $\frac{1}{2} \sum_{j=1}^5 \langle e_{i,T,j}^2 \rangle$.

Based on these considerations, the key elements are to use $f_{i,T,k} = \langle q_{i,T,k} \rangle$ and to estimate $\langle q_{i,T,k} \rangle$ with as little variability as possible. A potential approach is to make asset classes and to estimate the expected value as the observed average of the class over a time interval. In general, ETFs have lower volatility than stocks, which would make the first and fifth return quintiles more prevalent in stocks than in ETFs. Indeed, this behavior is observed within the M6 competition universe, with the salient exception of the VXX ETF (Fig. 1). Therefore, we consider stocks and ETFs as the main classes of assets.

Splitting the average of $q_{i,T,k}$ over one period of the M6 competition (four weeks) into $i \in \text{stocks}$ and $i \in \text{ETFs}$, just two main types of assets, leads to a fluctuating RPS that is generally lower than the benchmark (Fig. 2). This performance was assessed as the moving average of the RPS for 3 periods, a quarter of the M6 competition

duration, and 12 periods (48 weeks), the whole duration of the M6 competition. The values of $\langle q_{i,T,k} \rangle$ used in the calculations are approximately illustrated by Fig. 1 for $i \in \text{stocks}$ and $i \in \text{ETFs}$. They were computed as $\langle q_{i,T,k} \rangle_{C(i)} = \frac{1}{N_{C(i)}} \sum_{j \in C(i)} \sum_{T=T_l}^{T_F} q_{j,T,k}$ over the time interval from T_l to T_F , comprising the data available over the previous 50 periods. Here, $C(i)$ indicates the asset type of i , and $N_{C(i)}$ is the normalization factor. The details of the calculations are provided in the supplied code (<https://github.com/jmgvilar/M6competition>).

Another potential approach to estimating $\langle q_{i,T,k} \rangle$ is to consider a temporal average of each asset. The tradeoff is that short averaging intervals would lead to higher variability and longer ones would include outdated conditions. The approach considered averages over the last 5, 10, and 400 periods, weighted each by 0.2, 0.2, and 0.6, respectively. The results are generally better than the benchmark as well (Fig. 2). The average over the last 5 periods, from T_{5p} to T_F , was computed with the available data in that interval as $\langle q_{i,T,k} \rangle_{5p} = \frac{1}{N_{5p}} \sum_{T=T_{5p}}^{T_F} q_{i,T,k}$, where N_{5p} is the normalization factor. The averages over the last 10 and 400 periods, denoted by $\langle q_{i,T,k} \rangle_{10p}$ and $\langle q_{i,T,k} \rangle_{400p}$, were computed analogously. Consequently, the value of $\langle q_{i,T,k} \rangle$ was estimated as $\langle q_{i,T,k} \rangle_p = 0.2 \langle q_{i,T,k} \rangle_{5p} + 0.2 \langle q_{i,T,k} \rangle_{10p} + 0.6 \langle q_{i,T,k} \rangle_{400p}$.

The results can be improved further by estimating $\langle q_{i,T,k} \rangle$ as the average, $0.5 \langle q_{i,T,k} \rangle_{C(i)} + 0.5 \langle q_{i,T,k} \rangle_p$, of the asset-type and temporal averages (Fig. 2). This approach, with slightly different parameters in each submission, ranked first in the first quarter of the M6 competition and eighth globally out of 163 teams, with an RPS equal to 0.1573, which was very close to the best performer value of 0.1565 (Table 1).

It is important to emphasize that the asset-type, temporal, and combined averages, as computed over 48 weeks, consistently perform better than the benchmark across the entire five-year time-interval analyzed (Fig. 2). Specifically, the RPS for all these methods is consistently lower than the benchmark RPS at each time point, indicating superior predictive accuracy. This consistency in outperforming the benchmark over a prolonged time of five years strongly supports the robustness of these averaging methods in high-variability market conditions. In this context, Eq. (5) demonstrates that reducing forecast variability is crucial for improving predictive accuracy, particularly in high-variability conditions. This result suggests that grouping assets into classes, such as stocks and ETFs, can help minimize variability by capturing broader market trends while filtering out short-term noise. Fig. 2 confirms this empirically: The method of using two asset classes consistently leads to lower RPS compared to the benchmark, validating the effectiveness of this approach in reducing forecast variability and enhancing performance.

3. Investment decisions

The investment decisions consisted of selecting portfolios to obtain maximum returns adjusted by the risk. The evaluation was performed as detailed in Makridakis

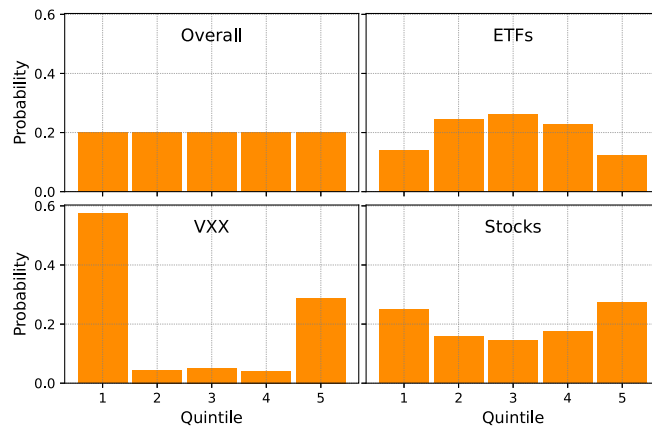


Fig. 1. Probability estimates of the returns of an asset type in each quintile. The histograms were generated from returns for the year 2022 for all the assets (top left), only ETFs (top right), only VXX ETF (bottom left), and only stocks (bottom right).

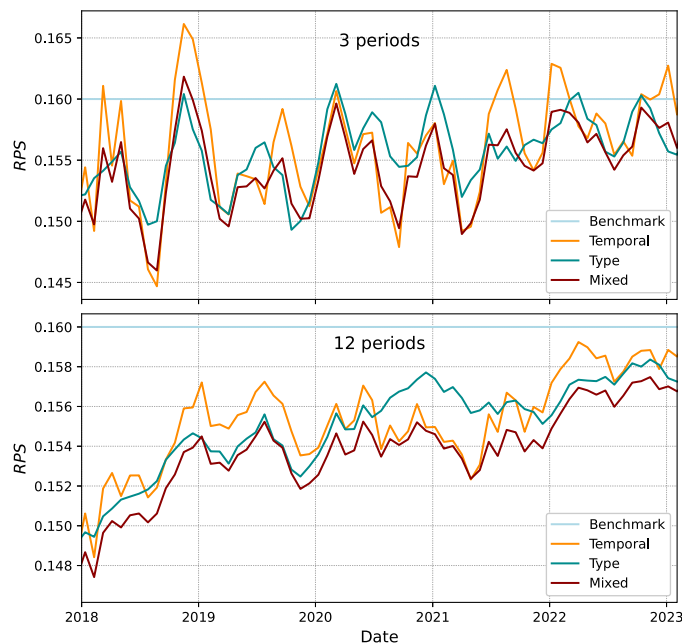


Fig. 2. Ranked probability scores (RPS) for different average procedures and moving time intervals. RPS values are computed for a moving time interval of 12 weeks (3 periods, top) and 48 weeks (12 periods, bottom) for different procedures used to estimate the probability of the ranks for each asset: overall probability estimate (benchmark), time average for each asset (temporal), separate estimates for stocks and ETFs (type), and combination of type and temporal averages (mixed).

Table 1

Key numerical results in the context of the M6 competition.

Metric	Best performer	Benchmark	Authors' approach
RPS (Q1)	0.15577	0.16000 (Rank: 49.5)	0.15577 (Rank: 1)
RPS (Global)	0.15645	0.16000 (Rank: 51)	0.15729 (Rank: 8)
IR (Q1)	11.5189	−0.05356 (Rank: 76)	2.53159 (Rank: 27)
IR (Global)	32.8898	0.45347 (Rank: 55)	1.30108 (Rank: 36)
Combined Rank (Q1)		38	3
Combined Rank (Global)		27	5

et al. (2023). In brief, the performance from day t_1 to t_2 is measured by

$$IR_{t_1:t_2} = \frac{ret_{t_1:t_2}}{sdp_{t_1:t_2}}, \quad (6)$$

where IR is referred to as the information ratio, ret represents continuously compounded returns, and sdp is the standard deviation of the daily returns. IR can be viewed as a variant of the conventional information ratio, but

with the benchmark return set equal to 0. It is also a variant of the Sharpe ratio with the risk-free rate set to 0.

The initial strategy consisted of selecting the stocks that performed better than half of the assets over 120 trading days. Additionally, stocks that performed better than 85% of the assets over 40 trading days were discarded. Namely, the strategy selects long-term performance but considers high short-term performance as a detrimental effect that would potentially be corrected to maintain the trend. This correction is analogous to the regression to the mean phenomenon, which indicates that if a random variable has an extreme value, the next value is likely to approach its mean (Stigler, 1997). The reversion can occur for various reasons, including overreaction or irrational market behavior. Investors who react to false information or make decisions too late may experience these mean-reverting movements as markets correct themselves.

In mathematical terms, to select the portfolio weights w_i , the approach considers the nonnormalized weights

$$v_{i,t} = \mathbf{I} \left\{ i \in \text{stocks} \wedge \text{rank} \left(\frac{S_{i,t-1}}{S_{i,t-121}} \right) > 50 \wedge \text{rank} \left(\frac{S_{i,t-1}}{S_{i,t-41}} \right) \leq 85 \right\}, \quad (7)$$

where $S_{i,t}$ represents the adjusted price of asset i at time t and $\mathbf{I}$ is the indicator function, which is 1 if its argument is true and 0 otherwise. Normalization leads to the portfolio weights

$$w_{i,t}^{\text{RtT}} = \frac{v_{i,t}}{\sum_i v_{i,t}}. \quad (8)$$

Here, the superscript RtT refers to the regression to the trend approach and should not be confused with the time variable. In this case, the ETF weights are 0.

In terms of returns, the regression-to-the-trend approach performs generally better than the benchmark, especially over long investment horizons (Fig. 3). In terms of the IR, the approach also performs better than the benchmark, although the improvement is not as marked (Fig. 4).

A subsequent strategy for improving the IR, not considered initially in the competition, was to compensate for fluctuations with short positions in ETFs. Given the nonnormalized weights for ETFs $u_i = \mathbf{I}\{i \in \text{ETFs}\}$, the overall weights of the portfolio were selected as

$$w_{i,t}^{\text{Compensated}} = \frac{2}{3} \cdot \frac{v_{i,t}}{\sum_i v_{i,t}} - \frac{1}{3} \cdot \frac{u_i}{\sum_i u_i}. \quad (9)$$

Namely, two-thirds of the portfolio was long positions in selected stocks and one-third consisted of short positions in ETFs. In this case, the ETF weights are $-1/(3 \times 50)$ for each of the ETFs. This approach, referred to as compensated, provided returns close to the benchmark (Fig. 3) but with IRs higher than those of both the benchmark and the regression to the trend approach (Fig. 4).

The choice of shorting just ETFs, rather than stocks or stocks and ETFs, was based on slightly better performance

over 48 periods (Fig. 5). In general, the performance of each shorting type over 12 or 3 periods was essentially the same, with only marginal fluctuating differences.

The regression to the trend approach ranked 27th the first quarter of the competition and switching afterward to the compensated approach ranked 36th globally out of 163 teams, with a global IR of 1.301, which was higher than the value of 0.4535 obtained by the benchmark. The combined performance of the approaches presented above ranked 3rd in the first quarter and 5th globally out of 163 teams (Table 1).

4. Discussion

The results of the M6 competition have shown that, over its 48-week overall duration, the majority of the teams underperformed the benchmarks: only 23.3% provided more precise forecasts, just 28.8% developed better portfolios, and only 6.7% achieved both of them (higher IR and RPS scores) (Makridakis et al., 2023). These results are analogous to the general tendency of most actively managed equity funds to underperform their benchmarks. The underperformance of actively managed funds has often been attributed to high management costs, limitations in high-volume trading, fees, and risk-managed portfolios, among other causes (Del Guercio & Reuter, 2014). These limitations were not present in the M6 competition and yet the results were aligned with those observed in actual trading settings.

Here, we have shown that predicting the expected values under high-uncertainty conditions, such as those assumed by the efficient market hypothesis, is more effective on average than trying to predict actual values. In general, there is a tradeoff between fine- and coarse-grained detail. Finer detail would produce more accurate results but also increase the variability of the average estimates. The information used in the approaches was restricted to the 100 assets of the M6 competition universe and the methodology was relatively straightforward (moving averages, asset-class averages, and ranking). Despite these inherent constraints, the approach outperformed both benchmarks and ranked fifth out of 163 teams in the global combined performance.

Whether investment decisions can benefit from forecasting the relative performance of multiple assets is an outstanding question. The results presented here show that forecasting performance is largely non-actionable because improvements over the naïve benchmark arise from symmetric quintile distributions, such as those from the 1st and 5th return quintiles, being more prominent in stocks than in ETFs but with only minor biases toward any of these quintiles.

Potential avenues to improving forecasting performance would need more sophisticated methods than averaging to estimate the expected values, such as Bayesian approaches to combine the joint average performance distribution across time for each specific asset with that of their class, whether stock or ETF. In this regard, rather than considering two classes of assets, it could still be beneficial to split stocks and ETFs into a few subclasses each, as in clustering approaches, and take baseline

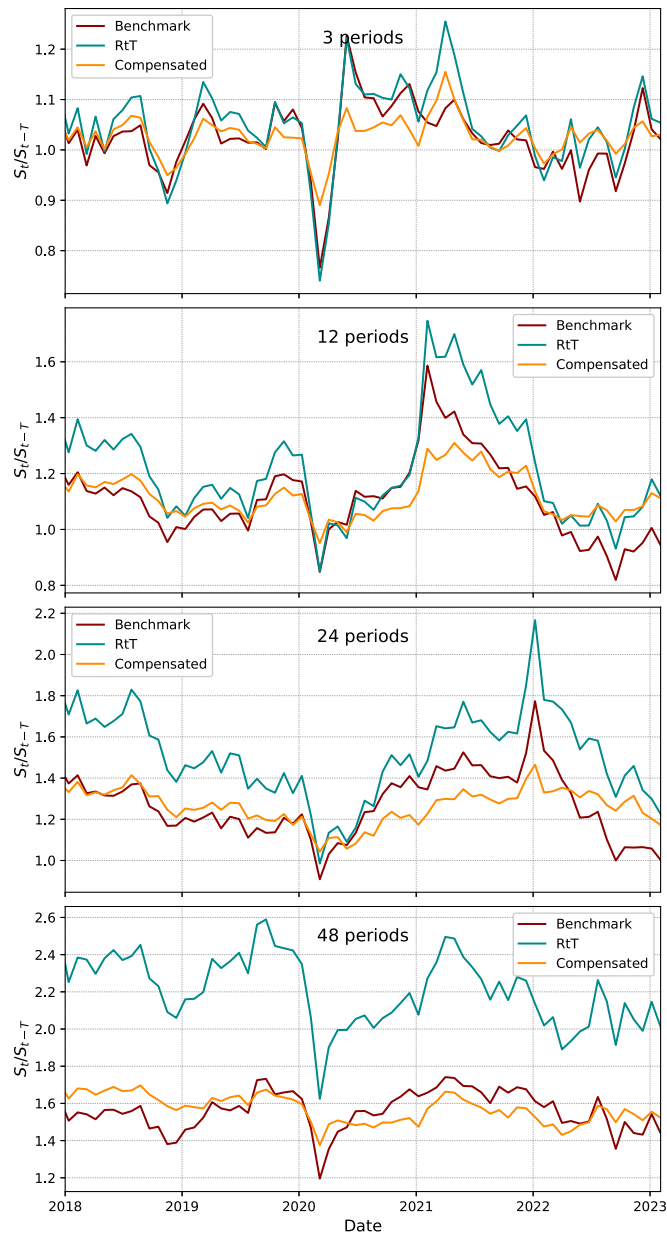


Fig. 3. Returns over 12 weeks (3 periods, top), 48 weeks (12 periods, second from top), 96 weeks (24 periods, second from bottom), and 192 weeks (48 periods, bottom) for the benchmark, return to the trend (RrT), and compensated portfolios.

averages within these subclasses. The minor differences observed among the RPS values of the top-performing teams (Makridakis et al., 2023), however, suggest that there may not be much room for improvement in this direction. In the case of investment decisions, a potential avenue for improvement would be to optimize long-term and short-term times individually for computing

each stock trend rather than uniformly selecting 120 and 40 days, respectively. We have included only information within the M6 competition universe. Therefore, a limiting aspect of the approach is the lack of available information, such as historical data for other asset prices, financial indicators, and other signals that could impact asset prices.

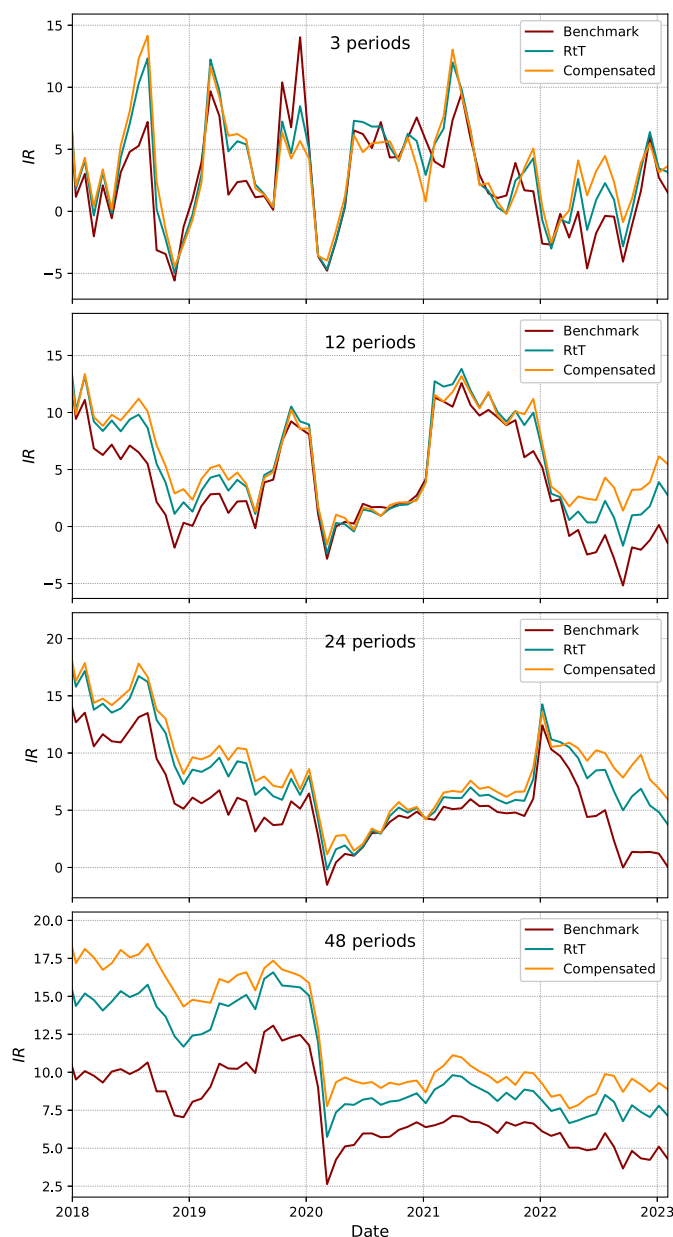


Fig. 4. Information ratio (IR) over 12 weeks (3 periods, top), 48 weeks (12 periods, second from top), 96 weeks (24 periods, second from bottom), and 192 weeks (48 periods, bottom) for the benchmark, return to the trend (RtT), and compensated portfolios.

CRediT authorship contribution statement

Jose M.G. Vilar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that

could have appeared to influence the work reported in this paper.

Data availability and code availability

A reproducibility package that includes all the necessary data and code to reproduce the results has been uploaded to GitHub as a python notebook and is available at <https://github.com/jmgvilar/M6competition>.

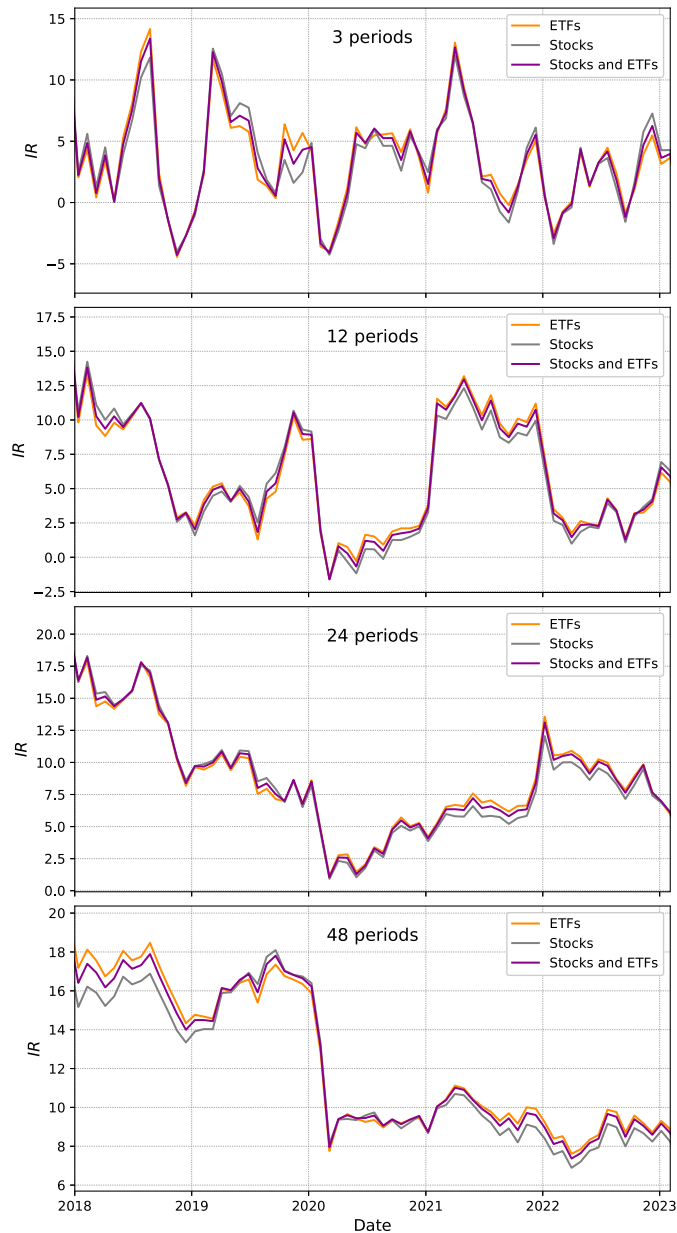


Fig. 5. Information ratio (IR) over 12 weeks (3 periods, top), 48 weeks (12 periods, second from top), 96 weeks (24 periods, second from bottom), and 192 weeks (48 periods, bottom) for compensated portfolios with just ETFs, as in Fig. 4, just stocks, and both stocks and ETFs.

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